

MSAJCE — Mohamed Sathak A.J. College of Engineering

Dept. of Artificial Intelligence & Machine Learning — Complete Data Extract for RAG Model

Source URL: <https://www.msajce-edu.in/aiml.php> | Sections: 14 sections covered

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1. Department Overview & Programme Details — B.E. AI & ML

Chunk ID: msajce_aiml_chunk_01 | URL: <https://www.msajce-edu.in/aiml.php#tab1> | ~430 tokens

Context: MSAJCE AIML department overview, B.E. programme details, seats, eligibility, and career prospects.

Artificial Intelligence (AI) is a technology that makes machines reproduce human behavior. Machine Learning (ML) is a sub-part of AI by which machines automatically learn from past data without programming. AI and ML together innovate human-like intelligent computer systems to solve complex problems.

B.Tech in CSE with Specialization in AIML (CSE-AIML) provides budding engineers with courses dedicated to frontiers in AI and ML with a foundation of Computer Science & Engineering. The curriculum covers Computational Mathematics, core Computer Science, machine learning, deep learning, natural language processing, computer vision, data mining, Explainable AI, Generative Adversarial Networks, Multimodal AI and Regenerative AI. Tools: Python, R, TensorFlow, Spark, Hadoop.

Career Prospects: Big Data Engineer, Business Intelligence Developer, Data Scientist, Machine Learning Engineer, Research Scientist, AI Data Analyst, AI Engineer, Robotics Scientist. Recruiting companies: Microsoft, Amazon, Goldman Sachs, Oracle GBU, Cisco, Dell Technologies, Accenture.

Programme Details

Detail	Value
Programme Name	B.E. Artificial Intelligence & Machine Learning (CSE-AIML)
Duration	4 years (Regular) / 3 years (Lateral Entry)
No. of Semesters	8 (Regular) / 6 (Lateral Entry)
Total Seats	60
Government Quota	30
Management Quota	30
Eligibility	10+2 with Physics, Chemistry and Mathematics
Curriculum	2021 Regulation — https://www.msajce-edu.in/uploads/academics/2021Regulation.pdf
Higher Studies	M.E. / M.Tech. / M.B.A.

Sample Questions this chunk answers:

- Does MSAJCE offer AI and ML course?
- What is the seat intake for AIML at MSAJCE?
- What is the eligibility for AIML at MSAJCE?
- How many years is the AIML programme?
- Which companies recruit AIML graduates from MSAJCE?

Keywords: AIML, artificial intelligence, machine learning, B.E. AIML, 60 seats, deep learning, NLP, Python, TensorFlow, Microsoft, Amazon, Goldman Sachs

2. Vision, Mission, PEOs, PSOs & Programme Outcomes (POs)

Chunk ID: msajce_aiml_chunk_02 | URL: <https://www.msajce-edu.in/aiml.php#tab2> | ~490 tokens

Context: AIML department Vision, Mission, Programme Educational Objectives, Program Specific Outcomes, and all 12 Programme Outcomes.

Vision

To be a centre of excellence for transforming students into proficient Artificial Intelligence and Machine Learning Engineers using through sustainable practices.

Mission

- M1. Impart core fundamental knowledge and necessary skills in Artificial Intelligence and Data Science through innovative teaching and learning methodology.
- M2. Inculcate critical thinking, ethics, lifelong learning and creativity needed for industry and society.
- M3. Cultivate the students with all-round competencies, for career, higher education and self-employability.

Programme Educational Objectives (PEOs)

PEO	Description
PEO1	Graduates will be prepared for analysing, designing, developing and testing software solutions and products with creativity and sustainability.
PEO2	Graduates will be skilled in the use of modern tools for critical problem solving and analyzing industrial and societal requirements.
PEO3	Graduates will be prepared with managerial and leadership skills for career and starting up own firms.

Program Specific Outcomes (PSOs)

PSO	Description
PSO1	Develop creative solutions by adapting emerging technologies/tools for real time applications of Industry.
PSO2	Apply the acquired knowledge to develop software solutions and innovative mobile apps for various automation applications.

Programme Outcomes (POs) — All 12

PO	Title	Description
PO 1	Engineering Knowledge	Apply knowledge of mathematics, science, engineering fundamentals and specialization to solve complex problems.
PO 2	Problem Analysis	Identify, formulate, review literature and analyze complex engineering problems.
PO 3	Design/Development	Design solutions for complex problems meeting specified needs with consideration for public health and safety.
PO 4	Investigations	Use research-based knowledge including design of experiments, analysis and interpretation of data.
PO 5	Modern Tool Usage	Create, select and apply appropriate techniques, resources and modern engineering/IT tools.
PO 6	Engineer & Society	Apply reasoning to assess societal, health, safety, legal and cultural issues.
PO 7	Environment & Sustainability	Understand impact of engineering solutions in societal and environmental contexts.
PO 8	Ethics	Apply ethical principles and commit to professional ethics and norms of engineering practice.
PO 9	Team Work	Function effectively as individual, member or leader in diverse teams and multidisciplinary settings.
PO 10	Communication	Communicate effectively, write reports, design documentation, make presentations.

PO 11	Project Management	Demonstrate knowledge of engineering and management principles as member and leader.
PO 12	Life-long Learning	Recognize need for and engage in independent and life-long learning in technological change.

Sample Questions:

- What is the vision of the AIML department at MSAJCE?
- What are the PEOs for AIML at MSAJCE?
- What are the programme outcomes for AIML?

Keywords: AIML vision, AIML mission, PEO, PSO, programme outcomes, OBE, engineering knowledge, ethics, sustainability

3. AI & ML Job Roles, Salary Trends & Employment Forecast India 2025-2030

Chunk ID: msajce_aiml_chunk_03 | URL: <https://www.msajce-edu.in/aiml.php#tab3> | ~520 tokens

Context: AI and ML employment forecast for India from 2025-2030, key job roles, required skills, and salary ranges.

The AI and ML sectors in India are poised for substantial growth over the next five years. Job Market Growth CAGR: 30-40% in next 5 years. Total AI jobs could double or triple. Key Drivers: Increased AI adoption across healthcare, finance, e-commerce, retail, manufacturing, agriculture. Government initiatives: Digital India, National AI Strategy, AI for All, AI-enabled Smart Cities. Global R&D centers: Google, Microsoft, IBM, Intel have set up R&D centers in India.

Salary Ranges — AI & ML Roles in India (2025–2030)

Experience Level	Annual Salary (INR)
Entry-level (0-2 years)	Rs. 6 – 12 Lakhs per annum
Mid-level (3-5 years)	Rs. 12 – 20 Lakhs per annum
Senior-level (5+ years)	Rs. 20 – 35 Lakhs per annum
AI Research Scientist	Rs. 25 – 50 Lakhs per annum
AI/ML Project Manager (Senior)	Rs. 30 – 60 Lakhs per annum

Key AI & ML Job Roles and Required Skills

Job Role	Key Skills	Demand Sectors
Machine Learning Engineer	Python, TensorFlow, PyTorch, Scikit-learn, neural networks	E-commerce, finance, automotive, manufacturing
Data Scientist	Python, R, SQL, Matplotlib, Seaborn, ML algorithms, big data	Finance, retail, healthcare, telecom
AI Research Scientist	Python, TensorFlow, PyTorch, mathematical modeling	Tech companies, universities, R&D labs
Deep Learning Engineer	TensorFlow, Keras, PyTorch, neural networks, computer vision	Healthcare, automotive, gaming, financial services
NLP Engineer	SpaCy, NLTK, sentiment analysis, speech recognition	Customer service, chatbots, voice assistants
AI Software Engineer	Python, C++, Java, TensorFlow, Docker, Kubernetes	Finance, retail, manufacturing, healthcare
AI Ethics Specialist	AI ethics, data privacy, algorithmic transparency	Government, big tech, research institutions
AI/ML Product Manager	Agile, data analysis, product lifecycle management	Tech companies, AI startups, enterprise software

Sample Questions:

- What is the salary of an AI engineer in India?
- What is the CAGR growth of AI industry in India?
- What salary can I expect after AIML degree?

4. Industry-Specific AI Trends, Certifications & Target Companies

Chunk ID: msajce_aiml_chunk_04 | URL: <https://www.msajce-edu.in/aiml.php#tab3> | ~440 tokens

Context: Industry-specific AI trends, certifications to upskill, and target companies for AIML job applications.

Industry-Specific AI and ML Job Trends

- **Healthcare:** Medical diagnostics, Personalized treatment, Drug discovery, Robot-assisted surgery, Bioinformatics, healthcare data analytics.
- **Finance and Banking:** Fraud detection, Credit scoring, Personalized banking, Automated trading. Fintech opportunities in Blockchain, smart contracts, AI in Cryptocurrency.
- **Automotive (Autonomous Vehicles):** Self-driving cars, Smart transportation. Skills needed: computer vision, sensor fusion, real-time decision-making.
- **Retail and E-commerce:** Personalized recommendations, dynamic pricing. Demand for recommendation algorithms and supply chain optimization.

Certifications and Skills to Upskill

No.	Certification / Skill
1	Python and R language for data manipulation and model development
2	TensorFlow, PyTorch, and Keras for deep learning
3	Pandas, NumPy, SciPy — data wrangling and analysis
4	SpaCy, Hugging Face — NLP frameworks for text processing and sentiment analysis
5	Google AI Certification & Deep Learning Specialization by Andrew Ng (Coursera)
6	Google Professional Machine Learning Engineer
7	AWS Certified Machine Learning — Specialty
8	Microsoft Certified: Azure AI Engineer Associate
9	IBM AI Engineering Professional Certificate
10	Data Science Professional Certificate
11	Certified Data Scientist (CDS)
12	Deep Learning Specialization by Andrew Ng
13	Advanced Machine Learning Specialization
14	Natural Language Processing with Deep Learning

Targeted Job Applications

AI-centric companies: Google, Microsoft, Amazon, NVIDIA, IBM, TCS, Infosys. Startups: India's growing AI/ML startups in Bangalore, Hyderabad, Pune. Freelancing: Upwork and Freelancer for AI/ML freelance opportunities.

Sample Questions:

- What certifications should I get for AI and ML?
- Is AWS ML certification useful after AIML degree?
- Does AIML have scope in healthcare?

Keywords: AI certification, AWS ML, Azure AI Engineer, Google ML Engineer, IBM AI, TensorFlow, Keras, Andrew Ng, Coursera, healthcare AI, fintech AI

5. AIML Faculty List & Laboratories

Chunk ID: msajce_aiml_chunk_05 | URL: <https://www.msajce-edu.in/aiml.php#tab1> | ~480 tokens

Context: All AIML department faculty with qualifications, designations, specializations, and lab details.

Faculty List

Name	Qualification	Designation	Area of Specialization
Dr. NAGASUBRAMANIAN R	M.E., Ph.D.	Head — PG	Computer Science Engineering
Dr. SUBRAMANIAN P	M.E., Ph.D.	Head — UG	Wireless Sensor Networking and Image Processing
Dr. PREMALATHA G	M.E., Ph.D.	Associate Professor	Image Processing and Machine Learning
Mrs. SARANYA V	M.Tech., (Ph.D)	Assistant Professor	IOT and Cloud Computing
Mrs. SASIKALA L	M.E.	Assistant Professor	Computer Science and Engineering
Mrs. KIRUTHIKA S	M.E.	Assistant Professor	Computer Science and Engineering
Mrs. ASRIN MAHMOOTHA A	M.E.	Assistant Professor	Computer Science Engineering
Mr. PRAVEENKUMAR P	M.E.	Assistant Professor	Computer Science Engineering
Mr. RAJAKUMAR B	M.Tech., (Ph.D)	Assistant Professor	IOT, AI, NLP and Image Processing
Mr. RAJASEKAR R	M.E., (Ph.D)	Assistant Professor	IOT, Network Security

Department Laboratories

No.	Lab Name	Course Code	Course Name	Virtual Lab Link
1	C Programming Lab	CS8261	C Programming	https://www.msajce-edu.in/images/departments/aids/Virtual-Lab-C-Programming.pdf
2	Internet Programming Lab	CS8661	Internet Programming	https://www.msajce-edu.in/images/departments/aids/Virtual-Lab-Internet-Programming.pdf
3	Operating Systems Lab	CS8461	Operating Systems	https://www.msajce-edu.in/images/departments/aids/Virtual-Lab-OS.pdf
4	OOAD Lab	CS8461	Object Oriented Analysis and Design	https://www.msajce-edu.in/images/departments/aids/Virtual-Lab-ood.pdf
5	Web Technology Lab	IT8511	Web Technology	https://www.msajce-edu.in/images/departments/aids/Virtual-Lab-Web-Technology.pdf
6	Networks Lab	CS8581	Networks Lab	https://www.msajce-edu.in/images/departments/aids/Networks.pdf

Sample Questions:

- Who is the HOD of AIML at MSAJCE?
- Who are the faculty in the AIML department?
- What labs are available in AIML department?

Keywords: AIML HOD, Dr Nagasubramanian, Dr Subramanian, Dr Premalatha, AIML faculty, IOT, image processing, NLP, network security, virtual lab

6. Course Materials — 2021 Regulation Subject List & Resource Links

Chunk ID: msajce_aiml_chunk_06 | URL: <https://www.msajce-edu.in/aiml.php#tab1> | ~460 tokens

Context: All AIML department 2021 Regulation subjects with codes and direct links to lesson plans, question banks and lecture notes.

S. No.	Code	Subject Name	Lesson Plan	Question Bank	Lecture Notes
1	MA3354	Discrete Mathematics	LP	QB	LN
2	CS3351	Digital Principles and Computer Organization	LP	QB	LN
3	AD3391	Database Design and Management	LP	QB	LN
4	AD3351	Design and Analysis of Algorithms	LP	QB	LN
5	AD3301	Data Exploration and Visualization	LP	QB	LN
6	AL3391	Artificial Intelligence	LP	QB	LN
7	AD3381	Database Design and Management Laboratory	LP	—	—
8	AD3311	Artificial Intelligence Laboratory	LP	—	—
9	GE3361	Professional Development	LP	—	—

Note: All resource PDFs are accessible at base URL: <https://www.msajce-edu.in/academics/aids/2021/> with subfolder LessonPlan/, QuestionBank/, LectureNote/, ICTTools/ and file naming convention: [SubjectCode]-LP.pdf, [SubjectCode]-QB.pdf, [SubjectCode]-LN.pdf, [SubjectCode]-ICT.pdf

Sample Questions:

- Where can I get AIML lecture notes at MSAJCE?
- What are the subjects in AIML 2021 regulation?
- What is the subject code for Artificial Intelligence?

Keywords: AIML course materials, 2021 regulation, MA3354, CS3351, AD3391, AD3351, AD3301, AL3391 Artificial Intelligence, lesson plan, question bank, lecture notes

7. AIML Placement Companies — Off-Campus Recruiters

Chunk ID: msajce_aiml_chunk_07 | URL: <https://www.msajce-edu.in/aiml.php> | ~360 tokens

Context: Companies that have recruited AIML and CSE students from MSAJCE through off-campus placements.

S.No.	Company (Group 1)	Company (Group 2)
1-4	Accenture	AT&T
5-8	BYJU'S	Capgemini
9-12	Carbonite	Chella Software
13-16	CISCO	Cognizant Technology Solutions (CTS)
17-20	Deloitte Consulting	Dell Technologies
21-24	EMIS Health	Future Focus Infotech
25-28	HCL Technologies	HP (Hewlett Packard)
29-32	IBM	Infosys
33-36	Infoview Technologies	Intellect Design Arena
37-40	L&T Infotech	Newage Software
41-44	Oracle	Saudi Airlines
45-48	Solverminds	Sutherland

49-52	TCS (Tata Consultancy Services)	Tech Mahindra
53-56	Wells Fargo	Wipro
57-60	Wolters Kluwer ELM	HTC

Sample Questions:

- Which companies recruit from MSAJCE AIML?
- Does TCS recruit from MSAJCE AIML?
- Does Wipro hire from MSAJCE?

Keywords: AIML placement, TCS, Wipro, Infosys, HCL, IBM, Accenture, Capgemini, Deloitte, CISCO, Oracle, Wells Fargo, Tech Mahindra, CTS, off-campus placement

8. Key Resource Links & General Contact

Chunk ID: msajce_aiml_chunk_08 | URL: <https://www.msajce-edu.in/aiml.php> | ~320 tokens

Context: All key download links, portal links, and general MSAJCE contact information relevant to the AIML department.

Resource	URL
2021 Regulation Curriculum	https://www.msajce-edu.in/uploads/academics/2021Regulation.pdf
Autonomous Regulation R24	https://www.msajce-edu.in/uploads/autonomous/AutonomousR24.pdf
AICTE Approval	https://www.msajce-edu.in/uploads/AICTE-APPROVAL.pdf
Anna University Approval	https://www.msajce-edu.in/uploads/AnnaUnivApproval.pdf
NAAC	https://www.msajce-edu.in/naac.php
IQAC	https://www.msajce-edu.in/iqac.php
NIRF	https://www.msajce-edu.in/nirf.php
ESE Timetable June 2025	https://www.msajce-edu.in/ESETT-June25.php
Exam Results	https://results.msajce-edu.in/
Online Admission	https://enrollonline.co.in/Registration/Apply/MSAJCE
Fee Payment	https://www.feepayr.com/
AICTE Scholarship	https://www.aicte.gov.in/schemes/students-development-schemes
Annual Report 2023-24	https://www.msajce-edu.in/uploads/aqar/AnnualReport2023-2024.pdf

Sample Questions:

- Where can I find the MSAJCE AIML syllabus?
- How to apply for admission to MSAJCE AIML?
- Where to check MSAJCE exam results?

Keywords: AIML syllabus, 2021 regulation, AICTE approval, NAAC, IQAC, NIRF, exam results, online admission, fee payment, annual report

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